

HS Data Science 26

Session 02: Scientific Working I — The Research Loop

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Session Overview

Module A — The Research Loop

Steps 1–11 of the scientific cycle:

- from reading literature to communicating results
- hypothesis as the contract with the reader
- three experiment modes: dev loop · pre-experiment · full

Module B — Experiment Design

- conditions, controls, and ablations
- choosing and applying the right metric
- pitfalls that invalidate results

Why this session?

Your seminar project is a *mini research loop*.

Every step applies — even at small scale.

The key insight: A well-designed experiment with a modest model beats a sophisticated model with a broken experiment every time.

Speaker notes

Open with the observation that most student project failures are not modelling failures — they are design failures: no clear question, no strong baseline, no pre-experiment. The session gives the vocabulary and structure to avoid that.



The Research Loop

Guiding Question: What steps turn a vague question into a publishable result?

The Research Loop — Overview

 The Research Loop — 11 numbered steps

Speaker notes

Walk through the 11 steps briefly. Stress two points: (1) iteration is not failure — the dashed arcs are normal science; (2) the steps before the experiment (1–5) determine quality more than the experiment itself.



The Scientific Research Loop as a cycle diagram

Speaker notes

The two dashed arcs are the most important part: "iterate: fix errors → re-run" and "revise hypothesis". Most real projects loop between steps 7–9 several times before reaching write-up.



Step 1 — Reading Literature

The three-pass method

(Keshav 2007)

Pass	What you read	Time
1st	title, abstract, conclusion	5 min
2nd	intro, headings, figures	20 min
3rd	full paper, related work	60–90 min

Only go to pass 3 for
directly relevant
papers.

While reading, ask:

- What claim does this paper make?
- What evidence supports it?
- What does it assume or leave out?
- What would a follow-up study do?



Active reading — example questions

- *"Could this result be a confound?"*
- *"What changes if the dataset is different?"*
- *"Does this generalise beyond the benchmark?"*

Keep a reading log:

one sentence per paper — the claim and why you care. Use Obsidian or Notion. Build this habit now.

Reading is not collecting — it is **understanding**
what is settled and **what remains open**.

Speaker notes

The three-pass method is from Keshav (2007) "How to Read a Paper." The reading log is essential — without it, students re-read papers they have already processed and lose the thread of related work.



Step 2 — Finding the Gap

Types of gap

(be complete — gaps fall into these categories)

Type	Example signal
Missing condition	"Only tested on English"
Conflicting results	"Papers A and B disagree"
Missing baseline	"No comparison to simple heuristic"
Scope limitation	"Offline only, not online"
Scalability limit	"Fails above 1M documents"
Methodological flaw	"No significance test"

Where to look:

-  "Limitations" and "future work" sections

A gap is not "nobody did this".

A gap is a *tension, a limit, or an open question that matters* — and that is investigable within your constraints.

From gap to research question:

"Does query reformulation improve answer quality in agentic search on the Open Web Index?"

The gap drives the question. The question drives the hypothesis. The hypothesis drives everything else.

- hedging language: *"we suspect"*, *"future work should"*
- points of disagreement across papers

Ask: *"What would a practitioner need that the literature does not yet provide?"*

Speaker notes

Students often confuse novelty ("nobody did this exact thing") with a real gap. The gap must be interesting AND bounded — investigable within a seminar timeframe. Walk through the gap-type table with a domain example from the group's topics.



Step 3 — The Research Question

From vague to sharp

Too vague	Sharp
"Is RAG better?"	"Does BM25 re-ranking improve answer faithfulness on TriviaQA vs. dense-only RAG?"
"Does my system work?"	"Does my system outperform a keyword-search baseline on OWI Curlie using nDCG@10?"

A sharp question names:
method · dataset · metric · comparison
.

Criteria for a good research question

- **Specific** — names method, dataset, and metric
- **Empirical** — requires evidence, not just reasoning
- **Interesting** — the answer changes how someone acts
- **Testable** — we can design an experiment to answer it
- **Feasible** — we have the resources and time to run it

Testable ≠ Feasible. *Testable*: can we construct the right experiment? *Feasible*: do we have the compute, data, and time? Both must be true.

Walk through the sharpening exercise. Ask each student to state their research question out loud, then challenge: "what dataset?", "what metric?", "compared to what?" The testable/feasible distinction is important — many students have a testable question but underestimate resources.

Step 4 — The Hypothesis

Hypothesis template:

"If [intervention], then [outcome], measured by [metric], on [data], compared to [baseline]."

Example:

"If we apply query reformulation before dense retrieval, nDCG@10 improves by $\geq 5\%$ vs. direct dense retrieval on OWI Curlie queries."

Falsifiability:

You must be able to run an experiment that proves you wrong

.



Properties to verify

Property	Check
Falsifiable	What result would disprove it?
Directional	Does it predict a specific direction?
Operationalisable	Can you measure it?
Feasible	Testable within your timeframe?
Grounded	Does related work make it plausible?

Anti-patterns

If no result could disprove the claim, it is a belief, not a hypothesis.

- "The system will be better" ← not directional
- "LLMs are intelligent" ← not falsifiable
- "Performance improves" ← no metric, no baseline

Speaker notes

The hypothesis is the contract between the researcher and the reader. If a student cannot state a falsifiable hypothesis, they are doing engineering exploration, not a scientific study. Both are valid — but they require different write-ups.



Step 5 — Experiment Design

Design before you code

Element	Decision to make
Conditions	which system variants to compare
Baseline	simplest reasonable alternative
Dataset	which data, which split, held-out test
Metric	primary + one secondary
Protocol	how many runs, how to average
Resources	GPU-hours, storage, API cost budget

Pre-registration

Write down hypothesis + design *before*

seeing results. This prevents:

- p-hacking
- HARKing (Hypothesising After Results are Known)
- post-hoc cherry-picking

Minimum for this seminar

1. One strong baseline
2. One primary condition
3. One ablation
4. Fixed train / dev / test split — no leakage
5. ≥ 3 runs with different seeds

A one-paragraph design document written before running is worth more than any metric

Check resources before committing. A full experiment that exceeds your compute budget is a wasted design.

improvement.

Speaker notes

The resource column in the design table is new and often missed by students. They design a full experiment and then discover on Day 5 that they need a GPU they don't have. Resource feasibility is a design decision.



Steps 6 & 7 — Three Experiment Modes

 Three Experiment Modes in AI/ML Research

Speaker notes

The three modes are not sequential stages — they are nested. The development loop runs inside the pre-experiment, which gates the full experiment. Students almost always skip the pre-experiment and pay for it: a week of full-experiment runs that reveal a broken pipeline.



Step 6 — Implementation



The development loop mindset

Work in small, verifiable steps:

1. code → run → inspect output
2. assert shapes, types, and ranges
3. overfit one batch deliberately — if you can't, the pipeline is broken
4. commit working state to Git after each milestone

Version control is not optional

```
feat: BM25 baseline on MSMARCO dev
```

```
MRR@10 = 0.187 - matches Thakur et al. 2021.  
Pipeline confirmed correct. Next: dense retrieval.
```

Environment and reproducibility

- pin dependencies (`requirements.txt` or `pyproject.toml`)
- fix random seeds at the top of every script
- keep data preprocessing in code — never edit raw data manually
- log every hyperparameter to a config file, not inline

Sanity checks before scaling up:

- Does random perform worse than your model?
- Does overfitting one batch produce near-zero loss?
- Do evaluation scores on dev match your expectation from related work?

Each commit message is a lab notebook entry.

Speaker notes

The overfit-one-batch test is the single most effective debugging technique in deep learning. If a model cannot memorise 32 examples, something in the architecture, loss, or data pipeline is wrong. Spend 30 minutes here before spending 3 days on a full run.

The pre-experiment is the **go / no-go gate**. Do not skip it.



Step 7 — Running Experiments · Step 8 — Results

Running experiments systematically

- one experiment = one folder with `config.yaml` + `results.json` + `README.md`
- log **all** runs — including the ones you will not report
- run every condition with ≥ 3 seeds
- measure wall-clock time and GPU memory — part of your resource budget

From raw numbers to reported results

1. Compute mean $\pm$ std across seeds
2. Compare conditions on the **held-out test set** — **once**
3. Do not re-run on test after seeing results
4. Plot distributions, not just means

Honest reporting checklist

- ☐ All conditions reported, not just the best
- ☐ Held-out test used only once
- ☐ Variance reported (not just mean)
- ☐ Baseline is as strong as you could make it
- ☐ No data leakage in the split
- ☐ Statistical significance tested

A smaller honest result is more valuable than a large result that cannot be reproduced.

The "test set used only once" rule is frequently violated. Once students see test results, they adjust their system and re-run. This is p-hacking, even if unintentional. Emphasise: dev set for tuning, test set for the final reported number only.

Step 9 — Ruling Out Errors

Common error types

Error	What it looks like
Data leakage	test items in training context
Wrong metric	metric $\neq$ what the hypothesis claims
Evaluation bug	off-by-one, wrong split, wrong average
Confound	model memorises dataset artefacts
Weak baseline	unfair comparison
Random variance	one lucky seed drives the result

Debugging process

1. Sanity-check: does random perform worse?
2. Overfit one batch on purpose
3. Inspect 20–30 model outputs manually
4. Ablate one component at a time
5. Re-run with 3 different seeds
6. Trace train/dev/test construction end-to-end

If results are surprisingly good: Stop and verify. Unexpectedly good results are more suspicious than bad ones.

Rule out errors *before* writing up — not during the



Error	What it looks like
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Distribution shift	train/test from different sources
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reviewer response.

Source: Pineau et al. 2021; Dodge et al. 2019

Speaker notes

"Surprisingly good results are suspicious" is a practical heuristic. Some of the most-cited retractions in ML came from pipeline bugs that looked like breakthroughs. Teach students to be genuinely sceptical of their own positive results.



Steps 10–11 — Write-up and Communicate

The paper is an argument, not a report

Section	Role
Abstract	claim + evidence + impact in 150 words
Introduction	motivation + gap + your contribution
Related Work	what exists and why it falls short
Method	what you did and why
Experiments	how you tested it
Results	what the evidence shows
Analysis	what it means and its limits
Conclusion	restate claim + future work

Write the abstract last

— once you know what you found.

Consolidation checklist

- ☐ One central claim — everything supports it
- ☐ Every figure caption states the takeaway
- ☐ Limitations section is honest
- ☐ Code and config available for reproduction

Communication formats

- **Talk:** 12-minute argument — one key idea
- **Poster:** one question, one answer, one figure
- **Report:** ACM format, max 8 pages

Say one thing clearly. Scientists remember papers that make a single sharp point.

Close the loop: after communication, other researchers read the paper, find the gap, and start their own cycle. The consolidation checklist maps directly to the NeurIPS reproducibility checklist — good preparation for research-track submissions.

Experiment Design in AI/ML

Guiding Question: How do you design an experiment that actually answers your research question?

Experiment Design — Overview

Topics in this module

- conditions, controls, and ablation studies
- evaluation metrics — choosing and applying them correctly
- common pitfalls that invalidate results
- statistical significance and effect size

Practical goal: After this module you can design a minimal but rigorous experiment for your seminar project — on paper, before writing a line of code.

Bridge from the loop overview to concrete mechanics. The module is deliberately focused on design, not implementation.

Conditions, Controls, and Ablations

Definitions

Term	Meaning
Control	unchanged reference — the baseline
Condition	a variant differing in one factor
Ablation	full system <i>minus</i> one component

Example — RAG pipeline

■ Baseline: BM25 + GPT-4

Design rule: one variable at a time

If Condition B differs from A in

both

retrieval method

and

prompt format, you cannot isolate the cause of any difference.

Change one thing at a time. When unavoidable, document the confound explicitly.

Minimum viable experiment

1. One baseline (simple, strong, cited)
2. One primary condition (your system)
3. One ablation (system minus your key component)

- Condition A: Dense retrieval + GPT-4
- Condition B: Dense + BM25 hybrid + GPT-4
- Ablation: B without re-ranking
- Ablation: B without query expansion

Each ablation isolates one contributing factor.

4. Two metrics (primary + secondary)

5. Fixed dataset split

Ablations are often more convincing than the main result — they show *what* drives the improvement.

Reviewers consistently ask for ablations. A paper with a system that outperforms baselines but has no ablation leaves open the question "but which part mattered?" — and reviewers will reject on that basis.

Metrics and Evaluation

The core principle

The metric must directly measure the property your hypothesis claims. Derive the metric from the hypothesis — never the reverse.

Common metrics by task
(representative, not exhaustive)

Task	Metric
Retrieval (ranked)	nDCG@k, MAP, Recall@k
Classification	F1 (macro/micro), MCC
Generation	ROUGE, BERTScore, BLEURT

Metric traps

Trap	Problem
BLEU for faithfulness	measures overlap, not factuality
Accuracy on imbalanced data	majority class trivially wins
Mean without variance	one lucky run inflates results
Proxy without validation	high proxy, low human preference

Dataset quality matters as much as the metric

Before using a benchmark, check:

- How was it constructed? By whom?
- Does it have known biases or artefacts?
- Are the labels reliable (inter-annotator agreement)?



Task	Metric
Faithfulness (RAG)	FactScore, Ragas faithfulness
Human eval	Likert, pairwise preference

A biased dataset with a correct metric still produces a misleading result.

Speaker notes

A common student mistake: choose whatever metric is easy to compute. Emphasise: metric follows from hypothesis. Also new: the dataset quality checklist. Many standard benchmarks have known biases (SNLI, SQuAD 2.0, etc.) — using them uncritically is a methodological error.



Common Pitfalls in ML Research

Data pitfalls

- **Train/test overlap** — shuffling before split leaks test items into training
- **Temporal leakage** — future data used to predict past events
- **Benchmark overfitting** — tuning on the test set, even unintentionally
- **Label noise** — gold labels wrong in 5–15% of NLP datasets
- **Dataset bias** — benchmark constructed from a specific source generalises poorly

Resource pitfalls

- **Insufficient compute** — full experiment requires GPU-hours not available
- **Storage underestimated** — embeddings, checkpoints, logs exceed quota

Model and reporting pitfalls

- **Spurious correlations** — model learns dataset artefacts, not the task
- **LLM-as-judge trained on same distribution** — evaluator is biased toward the model it judges
- **p-hacking** — try many metrics, report only the significant one
- **HARKing** — present post-hoc hypothesis as pre-specified
- **Selective ablations** — run 10, report 3
- **No error bars** — single run, no variance

Structural safeguards

1. Write hypothesis *before* seeing results
2. Commit evaluation script to Git *before* running on test set
3. Log every run — runs you don't report still happened

- **API cost overrun** — LLM calls at scale cost 10× the estimate

Estimate resources before running. A design that cannot be executed is not a design.

4. Report your worst result alongside your best

Pre-registration turns pitfalls into choices — you have to actively decide to violate them.

Source: Pineau et al. 2021; Dodge et al. 2019

Speaker notes

New in this version: resource pitfalls (compute, storage, API cost) and dataset bias/quality. These are the most frequent invisible failures in student projects. Walk through a concrete compute estimate: embedding 1M documents at 512-dim float32 = 2GB, plus index, plus checkpoints.



Statistical Significance and Effect Size

Statistical tests for ML

Situation	Test
Two systems, paired samples	Wilcoxon signed-rank
Two classifiers, same test set	McNemar's test
Means over multiple runs	Paired t-test ($n \geq 30$)
Multiple conditions vs. one baseline	Bonferroni correction

Threshold:

$p < 0.05$ conventional, but:

- report the exact p-value
- $p = 0.049$ and $p = 0.051$ are not meaningfully different
- apply multiple-test correction when comparing many conditions

Effect size matters more than p

Measure	Use
Cohen's d	standardised mean difference
Δ metric (absolute)	nDCG +0.03 vs. +0.003
Relative improvement	"+5% over baseline"

Practical significance check

Even a significant +0.001 nDCG is rarely worth deploying. Ask:

- Is it reproducible on other datasets?
- Is it consistent across subgroups?
- Would a user notice the difference?

Significant but tiny may matter less than non-significant but large and consistent.

Statistical significance is a necessary but not sufficient condition for a meaningful result. The practical significance questions bridge from statistical results to research conclusions. For seminar projects with small datasets, effect size is often more informative than p-values.

Wrap-Up

The Research Loop

- 11 steps: Literature → Gap → Research Q. → Hypothesis → Design → Implement → Experiment → Results → Analyse → Write-up → Communicate → repeat
- Iteration is not failure — it is the mechanism of science
- Steps 1–5 determine quality more than steps 6–8

Experiment Design

- Conditions, controls, ablations: one variable at a time



For your project — do this now:

1. Write your research question in one sentence (method · dataset · metric)
2. State a falsifiable, directional hypothesis
3. Name your baseline and justify it
4. Define your primary metric and why it matches your hypothesis
5. Estimate your resource budget (compute · data · time)

Next session (S03):

literature search in depth, reference management, the publication process, project management with GTD and Git.

Bring a written research question and hypothesis to S03. These will be reviewed as part of your

- Metric follows from hypothesis — never the reverse
- Pre-register: design before you see results
- Estimate resources before you commit to a design

research narrative (due 12/05).

Close by asking each student to write their research question and hypothesis right now — before leaving. Even a draft is better than nothing. This is the most effective 5-minute intervention in research methods teaching.

Image Credits & References

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